

Executive Summary: Statistical Testing Results

TikTok Claims Classification Project

Project Overview

The TikTok data team aims to develop a machine learning model to classify user submissions as claims. In this phase of the project, the team will perform a hypothesis test to examine the relationship between `verified_status` and `video_view_count`.

Key Insights

The analysis indicates a difference in view counts between TikTok videos posted by verified and unverified accounts. This suggests potential behavioral differences between the two account types.

Further investigation could explore the root cause of these differences. For example:

- Do unverified accounts tend to post more engaging videos, and are these videos claims or opinions?
- Are some unverified accounts associated with spam bots that artificially inflate view counts?

Details

The TikTok data team considered the relationship between `verified_status` and `video_view_count`.

One approach conducted was to examine the mean values of `video_view_count` for each group of `verified_status` in the sample data. The findings showed that unverified accounts have a mean of 265,663 views vs. 91,439 views for verified accounts

```
verified_status
not verified    265663.785339
verified       91439.164167
Name: video_view_count, dtype: float64
```

The second approach was a two-sample hypothesis test. Aligned with preliminary findings from the mean values, this statistical analysis shows that any observed difference in the sample data is due to an actual difference in the corresponding population means.

Next Steps

The team recommends proceeding with a regression model to predict `verified_status`. Analyzing verified status can provide insights into the behavior of verified users. These insights can then inform the interpretation of results from a subsequent claim classification model.